Hashing-Based Deep Metric Learning for the Classification of Hyperspectral and LiDAR Data

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***Abstract*— Multisource remote sensing data provide abundant and complementary information for land cover classification. Existing classification methods mainly focus on designing a multistream deep network to extract separate features of each single-source data, then adopting a fusing strategy to combine these extracted features for final classification. However, this kind of method neglects the sample correlation of single-source and cross-source data, which may deliver an unsatisfactory classification result when dealing with high intraclass-variability and low interclass-variability samples. To this end, a novel hashing-based deep metric learning (HDML) method is proposed for hyperspectral images (HSIs) and light detection and ranging (LiDAR) data classification in this article. First, a two-stream deep network is built to extract the spectral-spatial features of HSI and the elevation features of LiDAR, respectively. To fully use the complementary and correlated information of HSI and LiDAR data, we adopt attention-based feature fusion (AFF) modules to deliver a high-discrimination fused feature both for cross-source and single-source feature fusion. Then, the extracted features are fed into fully connected layers to generate class probabilities, respectively. Different from most existing methods that only utilize semantic information of samples, we elaborately designed a loss function to simultaneously consider the label-based semantic loss and hashing-based metric loss. Finally, a decision-level fusion strategy is adopted to further improve the classification results. Extensive experiments on three public HSI and LiDAR datasets demonstrate the effectiveness of the proposed method over some state-of-the-art approaches.**

***Index Terms*— Deep neural network, hashing learning, hyperspectral images (HSIs), light detection and ranging (LiDAR), metric learning, multisource data classification.**

I. Introduction

THE recent advances in sensor technology have made it possible to simultaneously acquire different source remote sensing data of the same investigated region. With the rich spectral information, hyperspectral images (HSIs) are widely used to discriminate the different materials [1], [2], [3], [4]. However, due to a lack the elevation information, it is difficult to distinguish the building roof and the road that is made up of concrete using HSIs. On the other hand, light detection and ranging (LiDAR) data record the elevation information about the surveyed area, which can be acquired even under cloud-cover conditions. Though LiDAR data can accurately classify land covers with different heights, it cannot discriminate roofs with different materials (e.g., asphalt and steel). Thus, the joint use of HSI and LiDAR data has been widely investigated in the remote sensing field to enhance the interpretation accuracy for the study area.

To fully use the complementary information between HSI and LiDAR data, a large number of works have been investigated in the past several years. Among them, feature-level fusion and decision-level fusion are two widely-used techniques. For the former one, the features of HSIs and LiDAR are separately extracted and then fused to improve the ability to representation of features [5]. However, the direct stacking of high-dimensional features may result in the curse of dimensionality and then lead to the Hughes phenomenon [6]. To address this problem, some subspace-based methods were proposed to fuse multisource features in a well-designed feature space. For instance, a sparse and low-rank technique was proposed for the fusion of HSI and LiDAR-derived features [7]. Liao et al. [8] proposed a generalized graph-based fusion method to couple dimension reduction and feature fusion for HSI and LiDAR classification. In [9], a multikernel learning strategy was also utilized to efficiently fuse the features of HSI and LiDAR. For decision-level fusion, there are also some successful applications. For instance, Hang et al. [10] adopted a weighted summation strategy to perform the decision-level fusion for the joint classification of HSI and LiDAR data. In [11], the linear and nonlinear features were fused in a decision phase.

Motivated by the successful progress of deep learning, HSI and LiDAR classification methods based on convolutional neural networks (CNNs) have emerged in the past several years. These methods can be divided into three categories in terms of network architecture: single-stream-network-based methods, two-stream-network-based methods, and three-stream-network-based methods. For the first one, the HSI and LiDAR data are first fused, and then the fused data is

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fed into a single-stream network to achieve feature extraction and fusion [12]. For the second category, a two-stream network is built to extract features of HSI and LiDAR, then the extracted features are fused in the subsequent fully connected layer [13], [14], [15]. More advanced, the three-stream network can excavate rich information from multisource data, where a two-stream network and a single-stream network are used to extract spatial-spectral features of HSI and spatial features of LiDAR, respectively [16], [17], [18]. Considering that the conventional convolution operation treats every pixel with equal weights during extracting features, which results in the reduction of classification performance, some researchers apply the attention mechanism to the classification task to focus on the area of interest and suppress the influence of unrelated information. Recently, a large number of attention mechanism-based methods have been proposed for the joint classification of HSI and LiDAR [16], [19], [20], [21], [22].

Although the aforementioned methods perform well in extracting discriminative features of each modality, there still exist limitations in the heterogeneous feature representation and complementary information fusion. To address this issue, some researchers have started to focus on information interaction between multiple modalities. Specifically, Zhang et al. [23] adopted a cross-attention module to exploit the complementary information, which enhances collaboration performance. Dong et al. [21] utilized a self- and cross-guided attention module to obtain refined spectral-spatial-elevation features and provide complementary information cross-guidance between HSI and LiDAR. In addition, other encode-decoded-based methods have also been developed to excavate the complementary information of HSI and LiDAR [15], [24]. This kind of method usually first obtains the fused feature of HSI and LiDAR via an encoder. Then, the corresponding data can be reconstructed via a decoder. Finally, by minimizing the error between the original data and reconstructed data, the fused features can preserve rich complementary information.

However, most existing methods focus on exploiting the semantic information of each sample and neglect the correlation between samples, which may deliver an unsatisfactory classification result when dealing with high intraclass-variability and low interclass-variability samples. To this end, we propose a novel hashing-based deep metric learning (HDML) method for HSI and LiDAR classification. On the one hand, we develop a two-stream attention-based network to simultaneously extract features of HSI and LiDAR. To fully use the complementary and correlated information between HSI and LiDAR data, attentional feature fusion (AFF) modules are utilized to deliver a high-discrimination feature both for cross-source and single-source feature fusion. On the other hand, we elaborately designed a loss function to simultaneously consider the semantic information of each sample and the correlation information between samples. Specifically, the label-based semantic loss and hashing-based metric loss are jointly utilized to constrain single-source and cross-source similarities, i.e., samples with the same classes should have a small feature distance and samples with different classes should have a large feature distance. In addition, we adopt a decision-level fusion strategy to further improve the classification results. The main contributions can be summarized as follows.

1) The features of HSI and LiDAR data extracted from a two-stream network are fused with multiple levels via AFF modules, which deliver a more discriminative feature representation.
2) A novel hashing-based similarity metric is used to excavate correlation information, which preserves the similarity information not only for single-source data but also for cross-source data.
3) The experimental results performed on three public datasets demonstrate that the proposed HDML outperforms other competitive methods in terms of classification accuracies.

The remaining of this article is organized as follows. Section II briefly introduces the related work. Section III describes the proposed method in detail. Comprehensive experiments and discussions are exhibited in Section IV. Finally, the conclusion is presented in Section V.

II. Related Work

A. Hashing Technique

The core idea of hashing is to learn a set of hash functions that map the high-dimensional feature vectors of original data, e.g., images, documents, and videos, into a set of low-dimensional binary codes in the Hamming space. A good hashing method is usually characterized by the following two properties. The first one is that the learned hash codes should be short to enable the easy storage of large amounts of data in memory. The second one is that the binary codes should preserve the similarity information of the original data as much as possible. Due to the above properties, hashing techniques have been widely used in similarity search [25], [26].

Existing hashing methods can be divided into data-independent methods and data-dependent methods. Data-independent methods randomly generate hash functions without using any training data. In contrast, data-dependent methods try to learn the hash functions by using some training data. In general, data-dependent methods can achieve better performance than data-independent methods with shorter hash codes. Depending on whether label information of training data is used, data-dependent methods can be roughly divided into two categories: unsupervised hashing and supervised hashing. For unsupervised hashing methods, the hash functions are learned from unlabeled training data. For supervised hashing methods, the supervised information is utilized to learn high-capacity hash codes, which are usually robust to noise and image transformations.

Recently, promoted by deep learning that has achieved great success in various computer vision tasks [27], deep hashing methods have been studied and achieved better performance than conventional hashing methods. For example, the hash learning process was decomposed into a stage of fitting approximate binary codes and a stage of simultaneously fine-tuning the image features and hash functions via a CNN [28]. In [29], pairwise label information was utilized to simultaneously learn image features and hash codes. By fully

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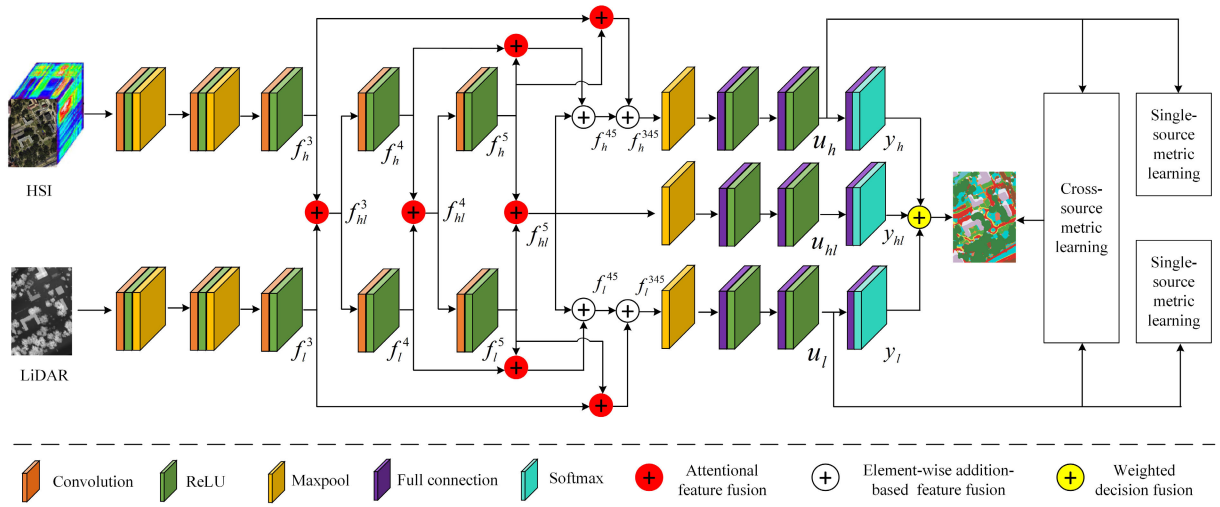


Fig. 1. Whole framework of the proposed HDML for HSI and LiDAR data fusion classification.

using deep networks and hashing techniques, deep hashing methods have become the mainstream method for similarity search.

B. Metric Learning in Remote Sensing

Metric learning, also known as similarity learning, aims to find an appropriate similarity measurement to preserve the distance structure that existed in the sample space. According to the adopted loss function, most of the existing metric learning works can be categorized into contrast loss-based metric learning and triplet loss-based metric learning. For the former category, the contrast loss function aims to minimize the similar pair distances and constrain dissimilar pair distances to be bigger than a margin. For the second category, the triplet loss function aims to constrain the negative pair distance to be larger than the positive pair distance plus a margin.

Recently, metric learning has been widely used in remote sensing image processing. For example, Cheng et al. [30] introduced metric learning into remote sensing image classification. Specifically, a metric learning regularization term was imposed on the CNN features to enforce models to be more discriminative. Zhang et al. [31] proposed an effective supervised metric learning algorithm for hyperspectral target detection. In such a method, the distance between the positive and negative samples was maximized by a designed objective function. Deng et al. [32] employed ideas from metric learning based on deep embedding features and made a similarity learning between pairs of samples. In addition, imposing the hashing technique into metric learning can further improve the performance of similarity learning. Thus, hashing-based metric learning methods have drawn increasing attention in recent years. In the field of remote sensing, hashing-based metric learning was adopted for HSI classification [33] and high-resolution scene retrieval [34], [35].

III. Proposed Method

Different from the recent works that usually neglect the rich correlation between samples, we aim to achieve HSI and LiDAR data fusion classification from a metric learning aspect. Specifically, a novel hashing-based similarity metric is designed to constrain feature distances between samples, i.e., samples with the same classes should have a small feature distance and samples with different classes should have a large feature distance. Fig. 1 shows the whole framework of the proposed HDML for HSI and LiDAR data fusion classification. In Sections III-A–III-C, we will introduce the proposed method in detail.

A. Feature Extraction and Fusion

Let $\mathbf{X}_h \in \mathbb{R}^{C\times M\times N}$ and $\mathbf{X}_l \in \mathbb{R}^{1\times M\times N}$ be an HSI and the corresponding LiDAR data, respectively, where $M \times N$ is the spatial size of image and C is the number of channels of HSI. As shown in Fig. 1, we build a two-stream network that consists of five convolutional layers and three fully connected layers to extract deep features. The data transmission of the first three convolutional layers is the same as the traditional convolutional layers, i.e., the input of the recent layer is the output of the previous layer. Considering that the original HSI and LiDAR have different shapes, the first convolutional layers of HSI-stream and LiDAR-stream networks adopt different convolutional kernels to process HSI and LiDAR data, respectively. The rest of the convolutional layers share the same network parameters, which arms to reduce the parameters of the whole network. Specifically, the outputs of the first three convolutional layers can be computed as follows:

$$\mathbf{f}_k^i = \begin{cases} \mathrm{Pool}\big(\mathrm{ReLU}\big(\mathrm{Conv}\big(\mathbf{f}_k^{i-1}\big)\big)\big), & i = 1, 2 \\ \mathrm{ReLU}\big(\mathrm{Conv}\big(\mathbf{f}_k^{i-1}\big)\big), & i = 3 \end{cases} \tag{1}$$

where $k \in \{h, l\}$ refers to the corresponding modality, Pool$(\cdot)$, ReLU$(\cdot)$, and Conv$(\cdot)$ denote the max pooling, Rectified Linear Unit function, and convolutional operations, respectively. $\mathbf{f}_k^0 = \mathbf{x}_k$ is image patch with the size of $p \times p$ extracted from the $\mathbf{X}_h$ or $\mathbf{X}_l$. Considering that multiple modalities exist in rich complementary and correlated information, we adopt multiple AFF modules to achieve cross-source feature fusion,

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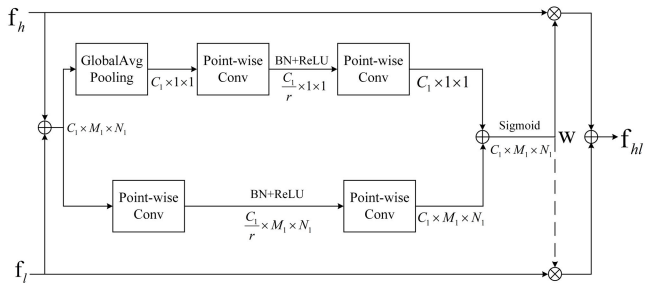


Fig. 2. Illustration of AFF module. $\oplus$ denotes the broadcasting addition and $\otimes$ denotes the element-wise multiplication.

which can improve information interaction between modalities. Specifically, we fuse the features of HSI and LiDAR data at the last three convolutions layers and the fused features are regarded as the inputs of the next layers. Through aggregating local and global contexts, the AFF module can benefit from an attention mechanism, which may overcome the semantic and scale inconsistency issue among input features [36]. Fig. 2 shows the architecture of an AFF module. Let $\mathbf{f}_h$ and $\mathbf{f}_l$ be input features with the shape of $C_1 \times M_1 \times N_1$. The fused feature $\mathbf{f}_{hl}$ can be expressed as

$$\mathbf{f}_{hl}^i = \mathrm{AFF}\left(\mathbf{f}_h^i, \mathbf{f}_l^i\right) = \mathbf{w} \otimes \mathbf{f}_h^i + (1 - \mathbf{w}) \otimes \mathbf{f}_l^i, \quad i = 4, 5 \tag{2}$$

where $\otimes$ denotes the element-wise multiplication, $\mathbf{w}$ denotes the attentional weight which can be computed by

$$\mathbf{w} = \sigma(\mathbf{L}(\mathbf{f}) \oplus \mathbf{g}(\mathbf{f})) \tag{3}$$

where $\mathbf{f} = \mathbf{f}_h \oplus \mathbf{f}_l$, $\oplus$ denotes the broadcasting addition. σ denotes the Sigmoid function. $\mathbf{g}(\mathbf{f})$ and $\mathbf{L}(\mathbf{f})$ denote global and local contexts, respectively. For the global context, it can be computed as

$$\mathbf{g}(\mathbf{f}) = B(\mathbf{W}_2 \mathrm{ReLU}(B(\mathbf{W}_{1\mathrm{GloPool}}(\mathbf{f})))) \tag{4}$$

where $B(\cdot)$ and GloPool$(\cdot)$ denote batch normalization and global average pooling, respectively. $\mathbf{W}_1 \in \mathbb{R}^{(C_1/r) \times C_1}$ is the parameter of a dimension reduction layer and $\mathbf{W}_2 \in \mathbb{R}^{C_1 \times (C_1/r)}$ is the parameter of a dimension increasing layer, where r is the channel reduction ratio. For the local context, it can be computed as

$$\mathbf{L}(\mathbf{f}) = B(\mathbf{W}_4 \mathrm{ReLU}(B(\mathbf{W}_3 \mathbf{f}))) \tag{5}$$

where $\mathbf{W}_3 \in \mathbb{R}^{(C_1/r) \times C_1 \times 1 \times 1}$ and $\mathbf{W}_4 \in \mathbb{R}^{C_1 \times (C_1/r) \times 1 \times 1}$ are the convolutional kernels.

In addition to the above cross-source feature fusion, we also apply AFF modules to each stream for single-source feature fusion. Specifically, the outputs of different layers of each stream are fused to improve the ability of representation. For example, for HSI-stream network, $\mathbf{f}_h^3$ and $\mathbf{f}_h^4$ are fused with $\mathbf{f}_h^5$ via an AFF module, respectively. Then, element-wise-based feature fusion is adopted to further fuse the above features. This procedure can be expressed as

$$\mathbf{f}_l^{345} = \mathbf{f}_l^{45} + \mathrm{AFF}\left(\mathbf{f}_l^3, \mathbf{f}_l^5\right) = \mathrm{AFF}\left(\mathbf{f}_l^4, \mathbf{f}_l^5\right) + \mathrm{AFF}\left(\mathbf{f}_l^3, \mathbf{f}_l^5\right) \tag{6}$$

where $\mathbf{f}_h^{45}$ and $\mathbf{f}_h^{345}$ denote the features that fuse the outputs of corresponding layers.

After that, the fused features, including $\mathbf{f}_h^{345}$, $\mathbf{f}_l^{345}$, and $\mathbf{f}_{hl}$ are fed into fully connected layers to further extract high-level semantic features and class probability distributions. Here, we denote $\mathbf{u}_h$, $\mathbf{u}_{hl}$, and $\mathbf{u}_l$ as the semantic features. $\mathbf{y}_h$, $\mathbf{y}_{hl}$, and $\mathbf{y}_l$ denote class probability distributions.

B. Objective Function Designing

After obtaining class probability distributions, we first adopt cross-entropy loss to minimize the error between the predicted labels and ground-truth labels, i.e.,

$$\mathcal{L}_{\mathrm{sem}} = -\frac{1}{K}\sum_{i=1}^{K}\left\langle \mathbf{t}_i, \log \mathbf{y}_{h_i} \right\rangle - \frac{1}{K}\sum_{i=1}^{K}\left\langle \mathbf{t}_i, \log \mathbf{y}_{l_i} \right\rangle - \frac{1}{K}\sum_{i=1}^{K}\left\langle \mathbf{t}_i, \log \mathbf{y}_{hl_i} \right\rangle \tag{7}$$

where $\langle\rangle$ represents inner production operation, K is the total number of samples, and $\mathbf{t}_i$ is the ground-truth vector of ith sample which is determined by the label of the centered pixel of the image patch. By minimizing the loss function $\mathcal{L}_{\mathrm{sem}}$, it can learn the discriminative semantic feature of each sample.

Different from most existing methods that only utilize semantic information to design objective functions, we also consider the similarity information to fully excavate the supervised information of samples. Specifically, we adopt hashing-based metric learning to constrain the feature distance both of cross-source samples and single-source samples. Let $\mathbf{u}_{h_i}$ and $\mathbf{u}_{l_j}$ be the semantic features of the ith HSI sample and the jth LiDAR sample, respectively. Then, the above semantic features are transferred to hash codes via binarizing, which can be formulated as

$$\begin{aligned} \mathbf{b}_{h_i} &= \mathrm{sgn}\left(\mathbf{u}_{h_i}\right), \quad i = 1, 2, \ldots, K \\ \mathbf{b}_{l_j} &= \mathrm{sgn}\left(\mathbf{u}_{l_j}\right), \quad t = 1, 2, \ldots, K \end{aligned} \tag{8}$$

where sgn$(\cdot)$ performs an element-wise operation for a matrix or a vector, i.e., sgn$(x) = 1$ if $x >= 0$ and -1 otherwise.

After binarizing, the hash code metrics can be represented as $\mathbf{B}_h = \{\mathbf{b}_{h_i}\}_{i=1}^K \in \mathbb{R}^{K \times L}$ and $\mathbf{B}_l = \{\mathbf{b}_{l_j}\}_{j=1}^K \in \mathbb{R}^{K \times L}$, where L is the length of hash codes. According to previous works that analyzed the effects of hash bits on performance [34], [35], we set the length to 64 in this article. Moreover, the likelihood of the cross-source pairwise labels $\mathbf{S}_{hl} = \{s_{ij}\}_{i,j=1}^K$ can be defined as

$$p\left(s_{ij} | \mathbf{B_h}, \mathbf{B_l}\right) = \begin{cases} \varphi\left(\omega_{ij}\right), & s_{ij} = 1 \\ 1 - \varphi\left(\omega_{ij}\right), & s_{ij} = 0 \end{cases} \tag{9}$$

where s_{ij} is the label of cross-source samples $(\mathbf{x}_{h_i}, \mathbf{x}_{l_j})$ such that $s_{ij} = 1$ if $\mathbf{x}_{h_i}$ and $\mathbf{x}_{l_j}$ come from same class and 0 otherwise. $\varphi(\cdot)$ is the logistic function and $\varphi(x) = (1/1 + e^{-x})$, $\omega_{ij} = (1/2)\mathbf{b}_{h_i}^T \mathbf{b}_{l_j}$. Based on the above definition, the cross-source metric loss can be given by taking the negative log-likelihood of the observed cross-source pairwise labels, i.e.,

$$\begin{aligned} \mathcal{L}_{\mathrm{cro}} &= -\log\ p(\mathbf{S}_{hl} | \mathbf{B}_h, \mathbf{B}_l) = -\sum_{s_{ij} \in \mathbf{S}_{hl}} \log\ p\left(s_{ij} | \mathbf{B}_h, \mathbf{B}_l\right) \\ &= -\sum_{s_{ij} \in \mathbf{S}_{hl}} \left(s_{ij}\omega_{ij} - \log(1 + e^{\omega_{ij}})\right). \end{aligned} \tag{10}$$

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Due to the discrete constraint in hash codes, it is hard to directly solve the problem (10). Motivated by [29] and [34], we can constrain the semantic feature $\mathbf{u}$ approach to hash code $\mathbf{b}$ as much as possible. Thus, the cross-source metric loss can be reformulated by

$$\mathcal{L}_{\text{cro}} = -\sum_{s_{ij}\in\mathbf{S}_{hl}} \left(s_{ij}\psi_{ij} - \log\left(1+e^{\psi_{ij}}\right)\right) \tag{11}$$

where $\psi_{ij} = (1/2)\mathbf{u}_{h_i}^T\mathbf{u}_{l_j}$. Through minimizing $\mathcal{L}_{\text{cro}}$, the feature distance in Hamming space (i.e., Hamming distance) between cross-source similar samples can be optimized to be as small as possible, and the Hamming distance between cross-source dissimilar samples becomes as large as possible.

In addition to the cross-source metric learning, we also perform the single-source metric learning on each source data. Similar to the above cross-source metric loss, we can formulate the single-source metric loss by

$$\begin{aligned}\mathcal{L}_{\text{sin}} &= \mathcal{L}_{h_{\text{sin}}} + \mathcal{L}_{l_{\text{sin}}} \\ &= -\sum_{s_{ij}\in\mathbf{S}_h}\left(s_{ij}\psi_{h_{ij}} - \log\left(1+e^{\psi_{h_{ij}}}\right)\right) \\ &\quad -\sum_{s_{ij}\in\mathbf{S}_l}\left(s_{ij}\psi_{l_{ij}} - \log\left(1+e^{\psi_{l_{ij}}}\right)\right)\end{aligned} \tag{12}$$

where $\mathcal{L}_{h_{\text{sin}}}$ and $\mathcal{L}_{l_{\text{sin}}}$ denote the single-source metric loss of HSI and LiDAR data, respectively. $\mathbf{S}_h$ and $\mathbf{S}_l$ are the labels of single-source samples. $\psi_{h_{ij}} = (1/2)\mathbf{u}_{h_i}^T\mathbf{u}_{h_j}$, $\psi_{l_{ij}} = (1/2)\mathbf{u}_{l_i}^T\mathbf{u}_{l_j}$.

Apart from the above cross-source and single-source metric constraints, quantitative loss is also defined to constrain the distance between semantic features and hash codes, which can be represented by

$$\mathcal{L}_{\text{qua}} = \sum_{i=1}^{K}\left\|\mathbf{u}_{h_i} - \mathbf{b}_{h_i}\right\|_2^2 + \sum_{j=1}^{K}\left\|\mathbf{u}_{l_j} - \mathbf{b}_{l_j}\right\|_2^2. \tag{13}$$

After that, the total loss function can be defined by simultaneously considering multiple losses including semantic loss, similarity loss, and quantitative loss, i.e.,

$$\mathcal{L} = \eta\mathcal{L}_{\text{sem}} + \alpha\mathcal{L}_{\text{cro}} + \gamma\mathcal{L}_{\text{sin}} + \lambda\mathcal{L}_{\text{qua}} \tag{14}$$

where η, α, γ, and λ are the regularization parameters to balance different constraints.

Finally, the objective function is defined by minimizing the above loss function, which can be expressed as

$$\begin{aligned}\mathcal{J} &= \min\mathcal{L} \\ &= \min\Bigg\{\eta\Bigg(-\frac{1}{K}\sum_{i=1}^{K}\big(\langle\mathbf{t}_i, \log\mathbf{y}_{h_i}\rangle \\ &\qquad +\langle\mathbf{t}_i, \log\mathbf{y}_{l_i}\rangle + \langle\mathbf{t}_i, \log\mathbf{y}_{hl_i}\rangle\big)\Bigg) \\ &\quad +\alpha\left(-\sum_{s_{ij}\in\mathbf{S}_{hl}}\left(s_{ij}\psi_{ij} - \log\left(1+e^{\psi_{ij}}\right)\right)\right) \\ &\quad +\gamma\Bigg(-\sum_{s_{ij}\in\mathbf{S}_h}\left(s_{ij}\psi_{h_{ij}} - \log\left(1+e^{\psi_{h_{ij}}}\right)\right) \\ &\quad -\sum_{s_{ij}\in\mathbf{S}_l}\left(s_{ij}\psi_{l_{ij}} - \log\left(1+e^{\psi_{l_{ij}}}\right)\right)\Bigg) \\ &\quad +\lambda\left(\sum_{i=1}^{K}\left(\left\|\mathbf{u}_{h_i} - \mathbf{b}_{h_i}\right\|_2^2 + \left\|\mathbf{u}_{l_i} - \mathbf{b}_{l_i}\right\|_2^2\right)\right)\Bigg\}.\end{aligned} \tag{15}$$

After obtaining the objective function, we can adopt the stochastic gradient descent (SGD) algorithm to learn all parameters.

C. Weighted Decision-Level Fusion

As mentioned above, we can obtain three kinds of class probability distributions, i.e., $\mathbf{y}_h$, $\mathbf{y}_{hl}$, and $\mathbf{y}_l$, via softmax function based on the corresponding semantic features. To further improve classification performance, an intuitive strategy is to fuse different classification results, which is also called decision-level fusion [37]. Existing decision-level fusion methods are mainly based on element-wise addition with the same weights, which may deliver a suboptimal fusion performance. To this end, we adopt a weighted decision-level fusion method to further optimize the classification result. Let $\mathbf{v}_h$, $\mathbf{v}_{hl}$, and $\mathbf{v}_l$ be the fusing weight vectors of $\mathbf{y}_h$, $\mathbf{y}_{hl}$, and $\mathbf{y}_l$, respectively. Note that these weight vectors have the same shape as the class probability distributions, i.e., $\mathbf{v}_k = \{v_{k_i}\}_{i=1}^{T} \in \mathbb{R}^{T\times 1}$, where T is the number of total classes, $k = \{h, hl, l\}$. The element of the weight vector can be computed by normalizing the number of training samples classified correctly, i.e.,

$$v_{k_i} = \frac{n_{k_i}}{n_{h_i} + n_{hl_i} + n_{l_i}}, \quad i = 1, 2, \ldots, T \tag{16}$$

where n_{k_i} is the number of training samples that are classified correctly for ith class.

Once the training is finished, we can obtain the weight vectors according to formulation (16). For the test pair $(\hat{\mathbf{x}}_h, \hat{\mathbf{x}}_l)$, we can obtain three class probability distributions $\hat{\mathbf{y}}_h$, $\hat{\mathbf{y}}_{hl}$, and $\hat{\mathbf{y}}_l$ via a feed-forward propagation. Then, the fused probability distribution can be computed by

$$\hat{\mathbf{y}} = \mathbf{v}_h \otimes \hat{\mathbf{y}}_h + \mathbf{v}_{hl} \otimes \hat{\mathbf{y}}_{hl} + \mathbf{v}_l \otimes \hat{\mathbf{y}}_l. \tag{17}$$

Finally, the label of the test sample can be determined by maximizing the probability, which can be expressed by

$$\text{Class}\left(\hat{\mathbf{x}}_h, \hat{\mathbf{x}}_l\right) = \arg\max_{i=1,2,\ldots,T}\hat{\mathbf{y}}_i. \tag{18}$$

IV. Experiments

In this section, we conduct comprehensive experiments on three widely-used HSI-LiDAR datasets to validate the effectiveness of the proposed HDML method.

A. Experimental Datasets

1) Houston Dataset: This dataset was collected by the NSF-funded National Center for Airborne Laser Mapping (NCALM) over the University of Houston campus and the neighboring urban area in June 2012, which was provided by the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Competition [38]. This dataset contains hyperspectral imagery and the corresponding

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TABLE I

DETAILED INFORMATION INCLUDING CLASS NAME, TRAINING, TEST SAMPLES OF HOUSTON, MUUFL, AND TRENTO DATASETS

Class No.	Houston dataset			MUUFL dataset			Trento dataset		
	Class Name	Training	Test	Class Name	Training	Test	Class Name	Training	Test
1	Healthy grass	198	1053	Trees	150	23096	Apple Trees	129	3905
2	Stressed grass	190	1064	Mostly Grass	150	4120	Buildings	125	2778
3	Synthetic grass	192	505	Mixed Ground	150	6732	Ground	105	374
4	Trees	188	1056	Dirt and Sand	150	1676	Woods	154	8969
5	Soil	186	1056	Road	150	6537	Vineyard	184	10317
6	Water	182	143	Water	150	316	Roads	122	3052
7	Residential	196	1072	Buildings Shadow	150	2083			
8	Commercial	191	1053	Buildings	150	6090			
9	Road	193	1059	Sidewalk	150	1235			
10	Highway	191	1036	Yellow Curb	150	33			
11	Railway	181	1054	Cloth Panels	150	119			
12	Parking Lot 1	192	1042						
13	Parking Lot 2	184	285						
14	Tennis Court	181	247						
15	Running Track	187	473						
	Total	2832	12197	Total	1650	52037	Total	819	29395

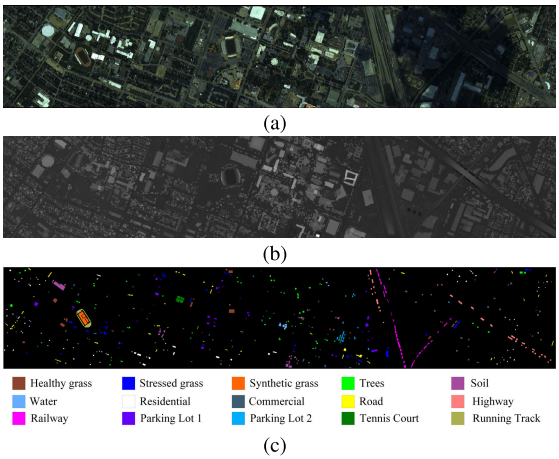


Fig. 3. Houston dataset. (a) Three-band false color composite. (b) Grayscale image of LiDAR-derived DSM. (c) Ground-truth map and color code.

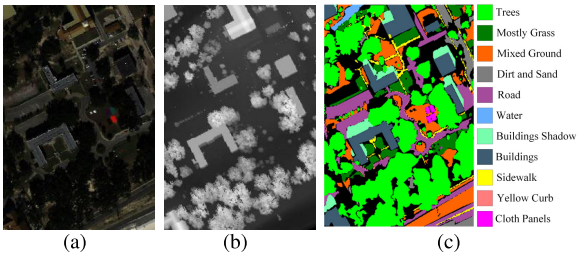


Fig. 4. MUUFL dataset. (a) Three-band false color composite. (b) Grayscale image of LiDAR-derived DSM. (c) Ground-truth map and color code.

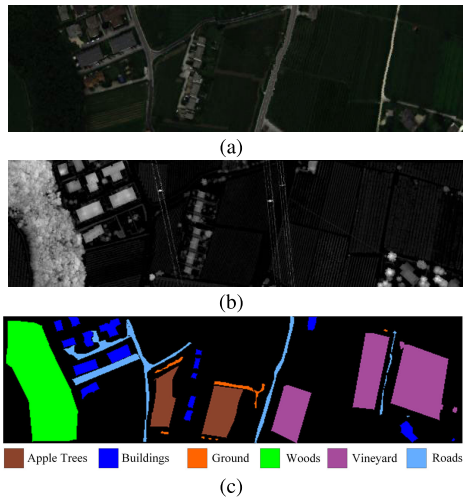


Fig. 5. Trento dataset. (a) Three-band false color composite. (b) Grayscale image of LiDAR-derived DSM. (c) Ground-truth map and color code.

registered LiDAR-derived DSM, both of which contain 349 × 1905 pixels with a spatial resolution of 2.5 m. The HSI has 144 spectral bands within the wavelength range from 380 to 1050 nm. There are 15 distinguishable land-cover classes in the studied scene. The left of Table I shows the detailed class names, training samples, and test samples. The false-color image of HSI, grayscale image of LiDAR-derived DSM, and ground-truth map are shown in Fig. 3.

2) MUUFL Dataset: The MUUFL dataset was collected over the campus of the University of Southern Mississippi Gulf Park, Long Beach Mississippi in November 2010. This dataset includes the registered HSI and LiDAR data, each of which has 325 × 220 pixels. The HSI has 64 spectral bands with wavelengths ranging from 375 to 1050 nm. The spatial resolution is 0.54 m across the track and 1.0 m along the track. There are 11 different classes in this scene. The detailed information including class names, training samples, and test samples is shown in the center part of Table I. In addition, Fig. 4 shows the false-color image of HSI, grayscale image of LiDAR-derived DSM, and ground-truth map of this dataset.

3) Trento Dataset: The third scene was captured over a rural area in the south of Trento, Italy. This size of the dataset is 166 × 600 pixels with a spatial resolution of 1 m. The HSI was collected by an airborne hyperspectral imaging systems (AISA) Eagle sensor with 63 spectral bands within a wavelength of range 0.42–0.99 m. The corresponding LiDAR data was acquired by an Optech Airborne Laser Topographic Mapping (ALTM) 3100EA sensor. There are six different types of ground objects in this scene. The right part of Table I shows the detailed class names, training samples, and test samples. The false-color image of HSI, grayscale image of LiDAR-derived DSM, and ground-truth map are shown in Fig. 5.

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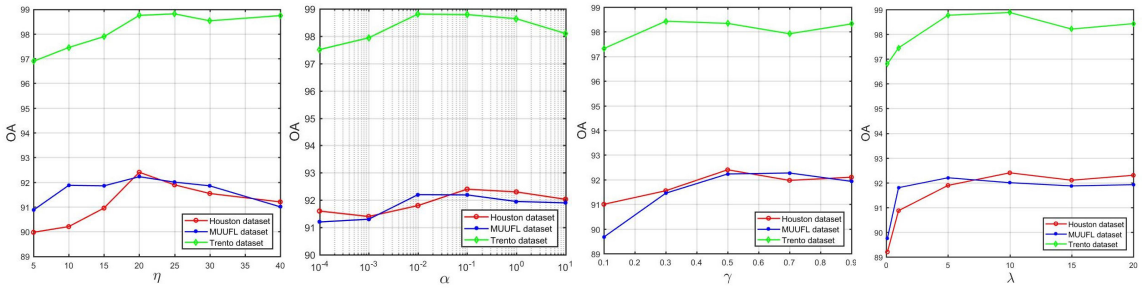


Fig. 6. Effects of different balanced parameters on OA. (a) Effect of η on OA. (b) Effect of α on OA. (c) Effect of γ on OA. (d) Effect of λ on OA.

B. Parameter Analysis

As shown in (15), there are four balanced parameters in our objective function, i.e., η, α, γ, and λ. Specifically, η is used to control the contribution of semantic information to the objective function. α and γ are used to control the contributions of cross-source and single-source similarity, respectively. λ constrains semantic features approach to hash codes. Apart from the balanced parameters, the size of patches p is another important parameter. Specifically, spatial information that existed in patches is useful for improving classification results [39]. In the following section, the effects of all the above parameters on classification results are individually analyzed. Note that when one parameter is analyzed, the other three parameters are set to fixed values.

1) Effects of η on OA: Fig. 6(a) shows the changing curve of the OA values with the increase of η on the three datasets. From this figure, one can see that the best OA values are achieved when $\eta = 20$ for Houston and MUUFL datasets. Although for the Trento dataset, the OA value under $\eta = 20$ does not reach the maximum value, the result at this time is still very close to the maximum value. Thus, we set $\eta = 20$ as the optimal value for the three datasets.

2) Effects of α on OA: Fig. 6(b) shows the changing curve of the OA values with the increase of α on the three datasets. As can be seen from this figure, when $\alpha < 0.01$, the OA values increase significantly and become stable with the increase of α. Through the above observation, it is found that the proposed method can obtain a satisfactory classification results on all datasets when α equals 0.1.

3) Effects of γ on OA: Fig. 6(c) shows the effects of γ on the OA values. From this figure, it is easy to see that the OA values significantly increase for all scenarios at the beginning. The main reason for the above experimental phenomenon is that the cross-source similarity information is utilized in the objective function, which may improve classification performance. In addition, we also observe that the OA values become stable with the increase of γ in the end. Based on the above observation, γ is set to 0.5 as the optimal value for the three datasets.

4) Effects of λ on OA: The corresponding changing curves of the OA values versus λ are shown in Fig. 6(d). As can be seen from this figure, one can see that the OA values reach the lowest value when λ is set to the smallest value. The main reason is that the objective function cannot effectively constrain the approximation error between the semantic features and the hash codes to be enough small. For both Houston and Trento datasets, the best OA values are achieved when $\lambda = 10$. Although for the MUUFL dataset, the OA values under $\lambda = 10$ do not reach the maximum values, the results at this time are still very close to the maximum values. Thus, we set $\lambda = 10$ as the optimal value for the three datasets.

TABLE II
EFFECTS OF THE SIZE OF PATCHES ON OA (PRESENTED BY %)

Datasets	7×7	9×9	11×11	13×13	15×15
Houston	91.13	91.74	**92.57**	92.02	91.68
MUUFL	90.82	91.49	92.20	**92.42**	92.05
Trento	96.98	97.45	98.78	**98.85**	98.65

5) Effects of p on OA: First, we adopt the above experimental results to set the values of four balanced parameters. Table II shows the OA values under different sizes on three datasets. As can be seen from this table, the OA values on the MUUFL and Trento datasets generally increase or become comparatively stable as the size becomes larger. By contrast, the influence on the Houston dataset is relatively sensitive. Specifically, the OA value first rises and then dramatically declines. The main reason is that the MUUFL and Trento datasets have larger smooth regions, and the Houston dataset has more detailed regions. In addition, we can easily see that the curves reach the best OA values when the sizes are set to 11×11, 13×13, and 13×13 for the Houston, MUUFL, and Trento datasets, respectively.

C. Classification Results

1) Quantitative and Qualitative Comparison: In order to validate the superiority of our proposed model in terms of classification performance, we systematically compare our method with some state-of-the-art methods, including CNN-based cross-channel reconstruction (CCR-Net) [15], two-branch CNN (TBCNN) [18], FusAtNet [40], Coupled-CNNs [10], deep encoder–decoder network (EndNet) [41], and hierarchical CNN-based transformer (HCT) [42]. In addition, we also perform classification experiments on the original HSI, LiDAR, and the fused HSI-LiDAR data, respectively, with support vector machines (SVMs) classifier [43]. We carefully reimplement the above-compared methods with the defeated settings in their papers. Note that the reimplemented results of some compared

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TABLE III
CLASSIFICATION RESULTS OBTAINED BY DIFFERENT METHODS ON THE HOUSTON DATASET. THE BEST VALUES ARE IN BOLDED

Class No.	SVM-LiDAR	SVM-HSI	SVM-HSI-LiDAR	EndNet	TBCNN	FusAtNet	CCR-Net	HCT	Coupled-CNNs	HDML
1	9.12	82.40	81.54	82.32	**83.10**	82.41	83.00	82.67	82.92	**83.10**
2	10.53	82.22	81.14	82.57	84.31	83.05	84.74	**93.40**	85.07	85.15
3	97.23	99.80	99.80	**99.93**	99.60	88.91	99.64	98.77	97.56	99.80
4	31.16	92.10	97.03	92.96	93.11	91.06	92.45	94.92	97.83	**99.34**
5	58.78	98.14	98.35	99.25	99.87	98.75	99.59	**100**	**100**	**100**
6	61.54	96.08	95.38	94.72	93.99	94.68	95.94	95.18	97.98	**98.60**
7	50.99	77.78	76.81	81.15	86.59	**92.71**	89.78	80.60	90.12	90.49
8	33.81	45.70	62.34	74.88	77.70	82.53	81.15	85.59	92.39	**95.44**
9	14.64	78.32	80.47	79.95	84.99	79.38	87.74	**88.84**	86.74	86.78
10	11.58	61.91	64.31	60.79	62.32	59.05	62.03	70.32	72.96	**76.06**
11	40.99	81.12	83.13	90.53	85.27	82.43	86.40	96.29	97.28	**98.20**
12	4.42	73.35	80.33	95.65	88.94	87.13	92.60	98.99	97.81	**99.14**
13	10.18	71.51	73.41	77.47	**91.16**	86.88	84.67	89.40	89.04	90.18
14	91.90	**100**	**100**	99.96	99.60	92.14	95.06	99.92	98.65	**100**
15	16.70	97.55	98.31	98.87	99.70	91.16	99.71	**99.83**	99.32	99.58
OA	30.56	79.60	82.46	85.55	86.42	84.75	87.36	90.28	91.21	**92.57**
AA	36.24	82.53	84.82	87.40	88.68	86.15	88.97	91.65	92.38	**93.45**
Kappa	25.49	78.06	81.08	84.37	85.31	83.48	86.33	89.44	90.47	**91.75**

TABLE IV
CLASSIFICATION RESULTS OBTAINED BY DIFFERENT METHODS ON THE MUUFL DATASET. THE BEST VALUES ARE IN BOLDED

Class No.	SVM-LiDAR	SVM-HSI	SVM-HSI-LiDAR	EndNet	TBCNN	FusAtNet	CCR-Net	HCT	Coupled-CNNs	HDML
1	47.82	83.18	85.53	86.30	91.60	**94.12**	90.79	88.84	93.43	93.66
2	42.03	83.67	84.93	82.82	86.98	85.31	80.91	86.58	87.04	**89.76**
3	15.95	70.06	70.22	71.46	75.01	70.88	72.94	79.86	82.85	**83.53**
4	42.66	86.37	85.20	90.20	95.16	90.93	90.43	94.56	95.33	**97.08**
5	77.28	88.69	88.22	90.00	90.71	86.98	89.19	85.01	90.71	**92.01**
6	98.94	92.40	99.05	99.34	99.56	99.15	99.68	99.64	99.93	**100**
7	33.21	89.85	90.62	90.84	**96.25**	92.49	93.14	95.06	94.17	94.91
8	66.76	78.65	90.59	92.06	94.66	95.34	95.69	93.54	96.77	**97.37**
9	21.84	73.47	74.29	81.15	84.11	79.75	77.56	76.53	**85.53**	82.43
10	26.26	96.36	94.95	97.27	97.58	94.95	93.03	95.08	94.28	**100**
11	83.47	98.15	98.32	98.74	98.15	95.24	99.08	98.95	98.88	**99.16**
OA	48.16	81.92	84.47	85.54	89.52	89.18	87.92	87.80	91.56	**92.42**
AA	50.57	85.53	87.45	89.11	91.80	89.56	89.31	90.33	92.63	**93.63**
Kappa	37.42	76.71	79.91	81.25	86.26	85.72	84.18	84.13	88.88	**89.74**

TABLE V
CLASSIFICATION RESULTS OBTAINED BY DIFFERENT METHODS ON THE TRENTO DATASET. THE BEST VALUES ARE IN BOLDED

Class No.	SVM-LiDAR	SVM-HSI	SVM-HSI-LiDAR	EndNet	TBCNN	FusAtNet	CCR-Net	HCT	Coupled-CNNs	HDML
1	1.30	89.39	89.75	94.11	99.48	98.90	**99.89**	98.57	98.95	99.00
2	44.73	82.75	97.17	99.36	98.28	96.09	98.79	**99.31**	96.71	97.77
3	0.00	96.74	96.52	94.29	74.44	82.09	82.25	95.72	93.48	**98.40**
4	64.13	96.47	99.34	**100.00**	99.41	98.71	99.98	99.94	99.89	**100**
5	87.02	78.47	82.63	98.80	99.94	97.07	95.09	75.70	**100**	**100**
6	80.06	68.12	89.20	**97.21**	84.06	81.76	88.18	86.31	90.65	92.10
OA	62.82	84.98	90.91	97.56	97.59	95.94	96.69	89.72	98.54	**98.85**
AA	46.21	85.32	92.44	97.30	92.60	92.44	94.03	92.59	96.88	**97.87**
Kappa	49.83	80.15	87.98	97.04	96.76	94.58	95.58	86.65	98.04	**98.40**

methods may be worse than the reported results of the original papers. To achieve a fair comparison, all data preprocessing techniques are excluded and the standard datasets are used for the compared experiments. For our proposed HDML, the size of patches p and four balanced parameters are set to optimal values according to Section IV-B. Apart from OA, other three metrics, i.e., average accuracy (AA), class accuracy (CA), and kappa coefficient, are also used to evaluate the classification performance.

Tables III–V list the quantitative comparison of different methods in terms of OA, AA, CA, and kappa on three experimental datasets. Several conclusions can be observed from these tables.

1) For the single-source methods, the classification results obtained by SVM-HSI are significantly better than the results obtained by SVM-LiDAR, which demonstrates that the spectral-spatial information existed in HSI is more important for discriminating different materials than elevation information existed in LiDAR.
2) By combining HSI with LiDAR data via a data-level fusion strategy, the classification results of SVM-HSI-LiDAR are further improved compared with SVM-HSI and SVM-LiDAR, which validates the superiority of multisource data utilization.
3) Except for SVM-based methods, other deep-learning-based methods all achieve satisfactory classification performance. The main reason is that feature-level fusion is performed via deep neural networks in these methods, which is better than traditional methods both in feature extraction and feature fusion.

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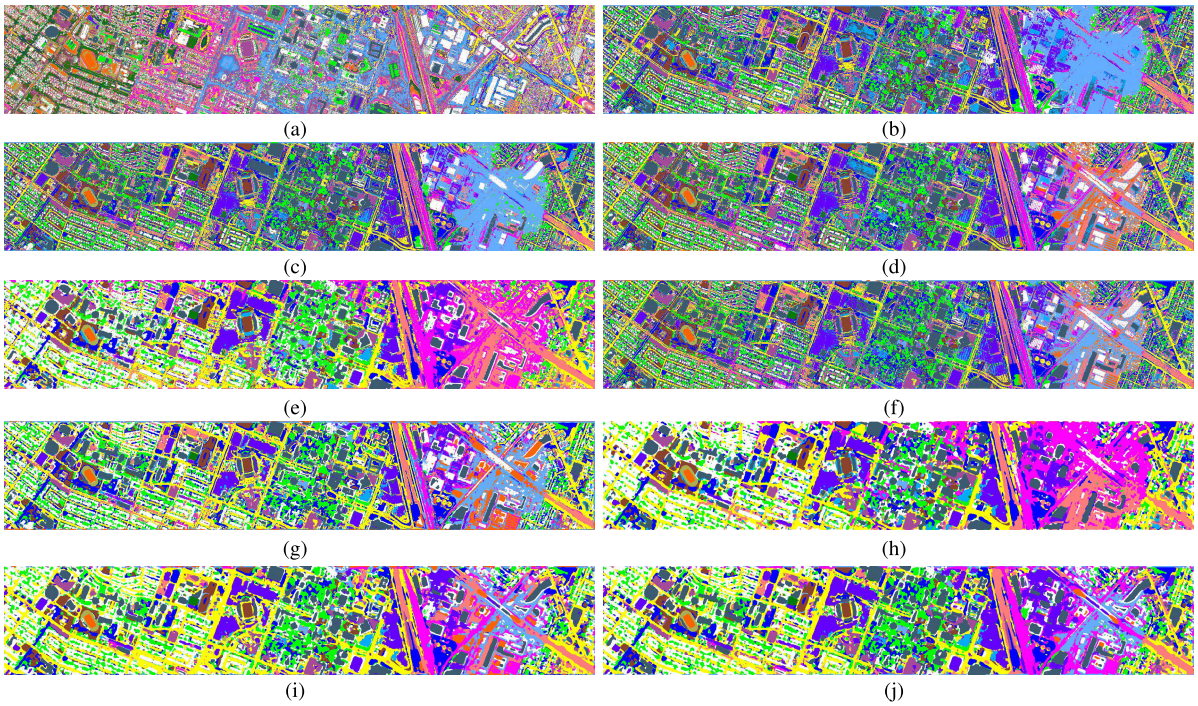


Fig. 7. Classification maps by different methods on the Houston dataset obtained by (a) SVM-LiDAR, (b) SVM-HSI, (c) SVM-HSI-LiDAR, (d) EndNet, (e) TBCNN, (f) FusAtNet, (g) CCR-Net, (h) HCT, (i) Coupled-CNNs, and (j) HDML.

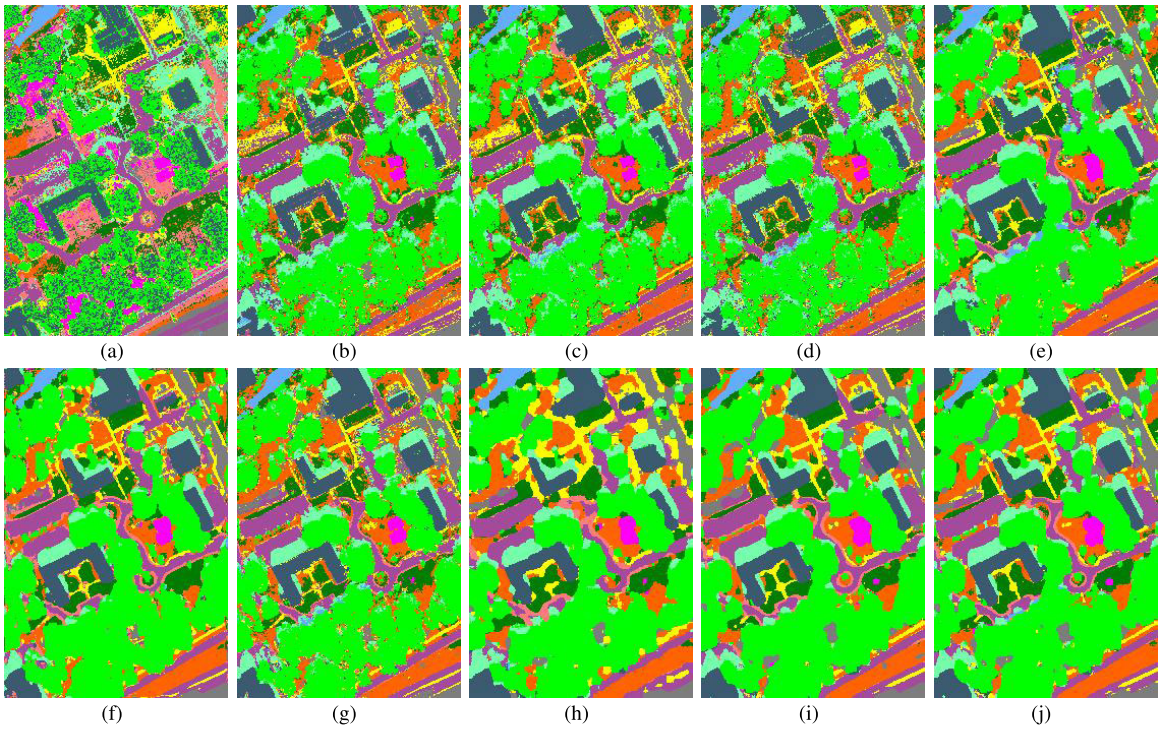


Fig. 8. Classification maps by different methods on the MUUFL dataset obtained by (a) SVM-LiDAR, (b) SVM-HSI, (c) SVM-HSI-LiDAR, (d) EndNet, (e) TBCNN, (f) FusAtNet, (g) CCR-Net, (h) HCT, (i) Coupled-CNNs, and (j) HDML.

4) Our method delivers the best classification results among all methods. Specifically, the OA values of our method obtain an improvement of 1.36%, 0.86%, and 0.31% than Coupled-CNNs on Houston, MUUFL, and Trento datasets, respectively. In addition, for classes with similar object materials, such as two different parking lots of the Houston dataset (i.e., Classes 12 and 13), our method achieved a significant improvement of 12.01% and 3.20% than FusAtNet.

For qualitative evaluation of the classification performance, visual maps are shown in Figs. 7–9. From these figures, one can see that SVM-LiDAR generates many outliers and

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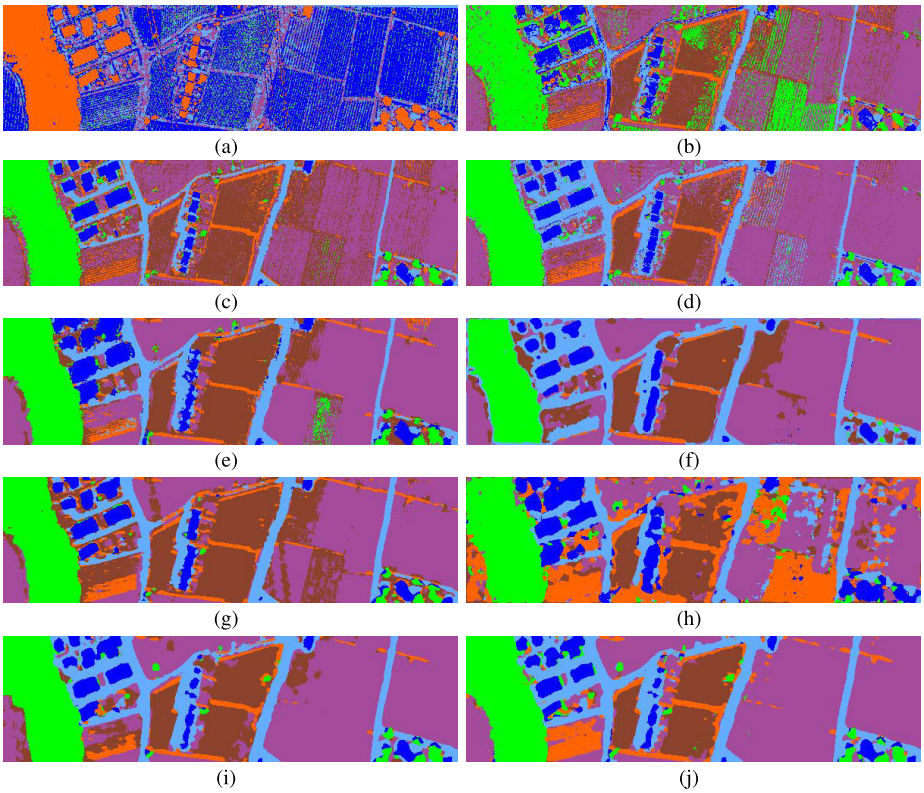


Fig. 9. Classification maps by different methods on the Trento dataset obtained by (a) SVM-LiDAR, (b) SVM-HSI, (c) SVM-HSI-LiDAR, (d) EndNet, (e) TBCNN, (f) FusAtNet, (g) CCR-Net, (h) HCT, (i) Coupled-CNNs, and (j) HDML.

misclassifies a lot of objects. By contrast, other methods obtain more homogeneous classification maps. Since all methods adopt image patches as the input, the classification maps are over-smoothed to some extent. Among all methods, the proposed method produces the most accurate classification maps with clearer boundaries when compared with the other methods, which is consistent with those in Tables III–V.

2) Feature Visualization: In this section, we have employed the t-stochastic neighbor embedding (t-SNE) algorithm to visualize the high-dimensional features. Specifically, we compare the proposed HDML with some deep-learning-based methods, including EndNet, TBCNN, FusAtNet, CCR-Net, HCT, and Coupled-CNNs. In addition, we also demonstrate the data distribution after directly fusing the original HSI and LiDAR data in spectral dimensionality. The visualization results of features in terms of t-SNE plots are shown in Fig. 10. As can be seen in this figure, the raw data points are difficult to distinguish. By contrast, the high-dimensional latent features extracted from some designed deep networks show good separability. In addition, compared with other competitors, the proposed method has better feature distribution with reasonable intracluster compactness and intercluster separability. The above experiments validate the effectiveness of the proposed method in learning discriminative features.

3) Comparison of Computational Complexity and Running Time: In this section, we have compared the proposed HDML with the other three competitors, including FusAtNet, HCT, and Coupled-CNNs, in terms of efficiency in the revised version. Table VI shows the running time (in seconds) and

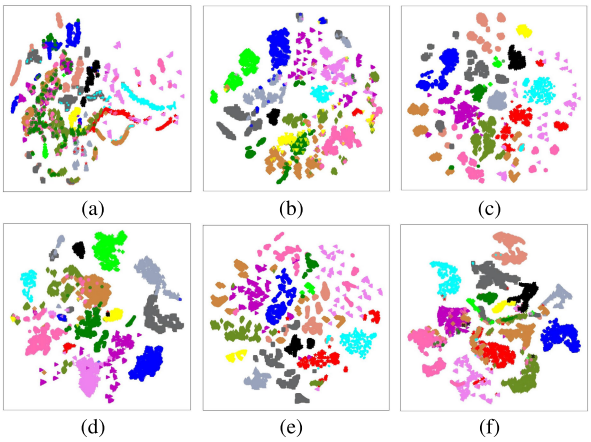


Fig. 10. t-SNE visualization results of (a) data points after fusing the raw HSI and LiDAR data, and the learned features in the latent space generated by (b) EndNet, (c) TBCNN, (d) CCR-Net, (e) Coupled-CNNs, and (f) HDML.

model complexity (in millions) of different methods on the three datasets, including the training and test time, the number of floating-point operations (FLOPs), and the number of trainable weight parameters. From this table, one can see that the FusAtNet has the longest running time and HCT has the fewest training time. For the proposed HDML, although the training is slightly higher than the ones of HCT and Couples-CNNs, the test time is the fewest among all methods, which indicates that HDML is more suitable for practical applications. In addition, HDML outperforms the compared methods in terms of complexity due to the fewer number of

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TABLE VI

COMPARISON OF RUNNING TIME (IN SECONDS) AND COMPLEXITY (IN MILLIONS) OF DIFFERENT METHODS

Methods	Metrics	Houston	MUUFL	Trento
FusAtNet	Train(s)	6780.56	6642.63	2357.34
	Test(s)	24.54	50.11	28.78
	FLOPS(m)	76.1417	74.8525	74.8169
	Params(m)	38.1092	37.4646	37.4468
HCT	Train(s)	123.32	74.19	38.56
	Test(s)	0.9524	4.05	2.34
	FLOPS(m)	7.9334	7.9800	7.9334
	Params(m)	0.4298	0.4304	0.4298
Coupled-CNNs	Train(s)	261.45	238.59	85.28
	Test(s)	2.70	9.10	2.73
	FLOPS(m)	2.2969	2.3302	2.2934
	Params(m)	0.1048	0.1035	0.1013
HDML	Train(s)	282.06	250.58	126.55
	Test(s)	0.50	1.92	1.12
	FLOPS(m)	2.0013	2.0353	1.9995
	Params(m)	0.0663	0.0658	0.0645

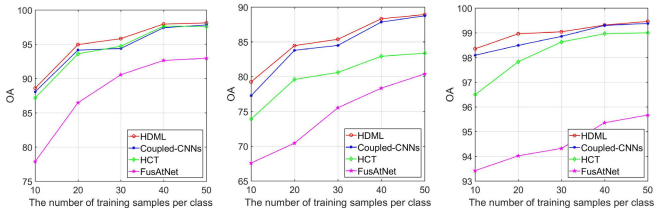


Fig. 11. Classification results in terms of OA values under different training samples on (a) Houston dataset, (b) MUUFL dataset, and (c) Trento dataset.

FLOPs and parameters, which also demonstrates the efficiency of the proposed method.

D. Classification Results Under Different Training Samples

To further verify the robustness and effectiveness of our proposed method, we conduct experiments to study the changing trend of OA values under the different circumstances of smaller training samples. Specifically, we randomly select a fixed number of samples from each category for training and the rest samples for test. The number of picks is increased from 10 to 50 in steps of 10. Apart from the proposed HDML, other three deep-learning-based methods are also considered for comparison, including FusAtNet, HCT, and Coupled-CNNs. Fig. 11 shows the performance sensitivity to the volumes of training samples. It can be seen from this figure that the classification accuracies of all the methods improve with the increase in the number of training samples, which validates the positive impact of training samples on the supervised deep-learning-based methods. In addition, the proposed HDML always achieves the best performance under all circumstances of sample selections.

E. Ablation Study

The proposed method benefits from three kinds of fusion strategy, including cross-source AFF (denoted as CS-AFF), single-source AFF (denoted as SS-AFF), and weight decision fusion (WDF). Specifically, CS-AFF is used to explore the interaction information between HSI and LiDAR data. SS-AFF is used to extract the discriminative features from HSI and LiDAR data, respectively. WDF is further adopted to combine multiple classification results. The contributions of each fusion strategy are evaluated by iteratively removing each of these strategies. As listed in Table VII, the classification accuracies in terms of OA values of the proposed HDML and its five variants are reported, where × and ✓ refer to without and with the corresponding strategies, respectively. From this table, several conclusions are observed. First, by comparing HDML_v1 and HDML_v2 or HDML_v5 and HDML, the WDF indeed improves the classification performance. Specifically, taking the Houston dataset as an example, the OA values obtain the improvements of 0.88% and 0.43% for the above two comparisons. Then, by comparing HDML_v3 and HDML_v4, it is easy to see that CS-AFF makes more contributions to classification performance than SS-AFF, which indicates that exploring interaction information between modalities is very important for multisource classification. Furthermore, by simultaneously using all fusion strategies, the proposed HDML delivers the best results on three datasets.

TABLE VII

ABLATION STUDY OF DIFFERENT FUSION STRATEGIES IN TERMS OF OA (PRESENTED BY %)

Methods	Fusion Strategies			Datasets		
	CS-AFF	SS-AFF	WDF	Houston	MUUFL	Trento
HDML_v1	×	×	×	88.73	89.34	96.03
HDML_v2	×	×	✓	89.61	90.41	96.68
HDML_v3	×	✓	✓	90.57	91.02	97.05
HDML_v4	✓	×	✓	91.73	91.64	97.91
HDML_v5	✓	✓	×	92.14	91.97	98.51
HDML	✓	✓	✓	**92.57**	**92.42**	**98.85**

V. CONCLUSION

In this article, we propose a novel multisource remote sensing data classification method, called HDML, by developing a deep hashing-based metric learning module. Different from existing methods, the proposed method makes full use of the complementary and correlated information of HSI and LiDAR data from the perspective of metric learning. In addition, multiple fusion strategies are adopted to further improve the discriminability of features. Finally, experiments on three datasets demonstrate the superiority compared with some state-of-art methods.

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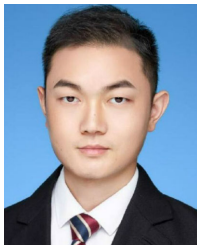
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